DO NOW — RECALL

Search and sort: comparisons and Big O

Name: _____ Class: _____ Date: _____

1. What property does binary search need that linear search does not?

2. What time complexity is typical of merge sort?

3. For n=5, how many times does a single for-loop from 0 to n-1 run?

4. A nested loop runs 4 times outside and 4 times inside. How many inner operations occur?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Sorted input.

2. Model answer

O(n log n).

3. Model answer

5 times.

4. Model answer

16 operations.

Vocabulary — know these terms

time complexity — How an algorithm's work grows with input size.

O(n) — Work proportional to input count.

O(log n) — Work proportional to the number of halvings.

$O(n^2)$ — Quadratic growth, often nested loops.

trade-off — A benefit balanced against another cost.

NEW LEARNING

Read, annotate and explain

Count what grows

Big O describes how work increases with input size, not the exact milliseconds on one computer.

Compare search

Linear search can take n checks; binary search on sorted input roughly halves the remaining candidates each step.

Compare sorting

A simple nested comparison loop can be $O(n^2)$, whereas merge sort is $O(n \log n)$.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

For 16 values, compare maximum comparisons of linear and binary search.

Predict first

Assume one successful or unsuccessful lookup in sorted data.

Model solution

Linear search can inspect all 16 elements; binary search checks at most 5 midpoints using this inclusive implementation.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name the complexity of a single loop visiting n values.

Question 2

An array has 128 ordered items. Compare linear and binary worst-case checks.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Is binary search always better than linear search? Give one counterexample.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

O(log n) means execution time is literally log n seconds.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.